1.4 A current of 7.4 A flows through a conductor. Calculate how much charge passes through any cross-section of the conductor in 20 seconds.

1.6 The charge entering a certain element is shown in **Fig. 1.23**. Find the current at:

(a) $t = 1$ ms

(b) $t = 6$ ms

(c) $t = 10$ ms

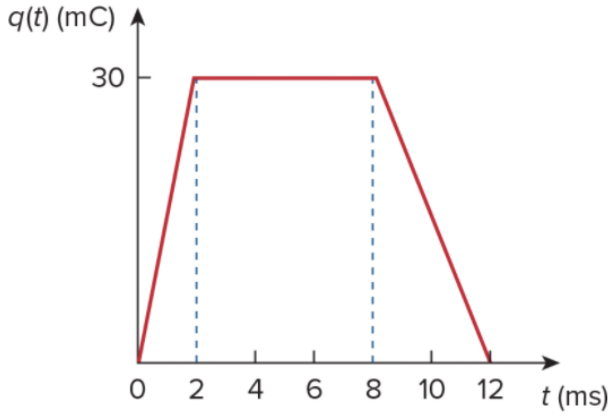

Figure 1.23
For **Prob. 1.6**.

1.8 The current flowing past a point in a device is shown in **Fig. 1.25**. Calculate the total charge through the point.

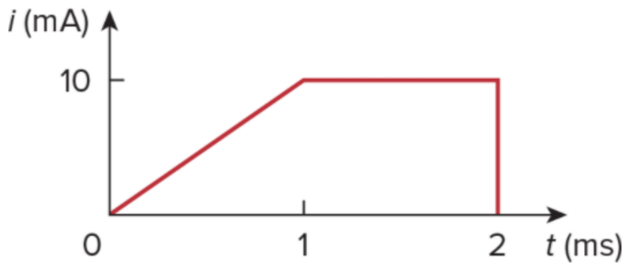

Figure 1.25
For **Prob. 1.8**.

1.11 A rechargeable flashlight battery is capable of delivering 90 mA for about 12 h. How much charge can it release at that rate? If its terminal voltage is 1.5 V, how much energy can the battery deliver?

1.12 If the current flowing through an element is given by

$$i(t) = \begin{cases} 3t\text{A}, & 0 \leqq t < 6\text{ s} \\ 18\text{A}, & 6 \leqq t < 10\text{ s} \\ -12\text{A}, & 10 \leqq t < 15\text{ s} \\ 0, & t \geqq 15\text{ s} \end{cases}$$

Plot the charge stored in the element over $0 < t < 20$ s.

1.16 Figure 1.27 shows the current through and the voltage across an element.

(a) Sketch the power delivered to the element for $t > 0$.

(b) Find the total energy absorbed by the element for the period of $0 < t < 4$s.

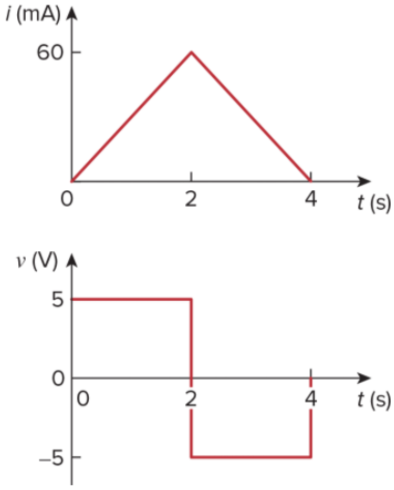

Figure 1.27
For Prob. 1.16.

1.18 Find the power absorbed by each of the elements in Fig. 1.29.

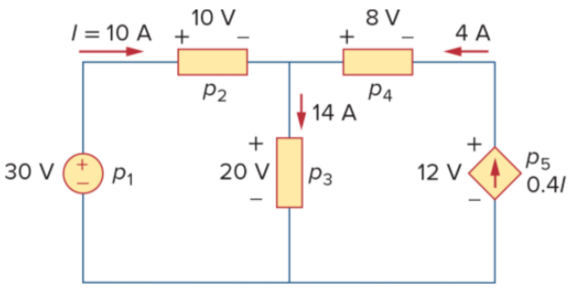

Figure 1.29
For Prob. 1.18.

1.20 Find V_o and the power absorbed by each element in the circuit of Fig. 1.31.

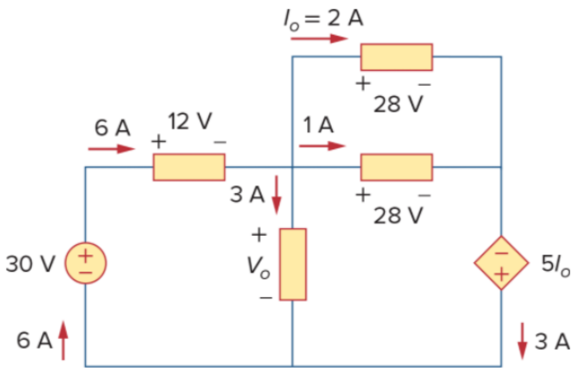

Figure 1.31
For Prob. 1.20.

1.24 A utility company charges 8.2 cents/kWh. If a consumer operates a 60-W light bulb continuously for one day, how much is the consumer charged?

1.28 A 60-W incandescent lamp is connected to a 120-V source and is left burning continuously in an otherwise dark staircase. Determine:

(a) the current through the lamp,

(b) the cost of operating the light for one non-leap year if electricity costs 9.5 cents per kWh.

2.6 In the network graph shown in **Fig. 2.70**, determine the number of branches and nodes.

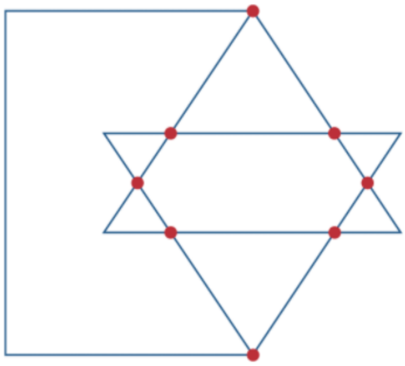

Figure 2.70
For **Prob. 2.6**.

2.7 Determine the number of branches and nodes in the circuit of **Fig. 2.71**.

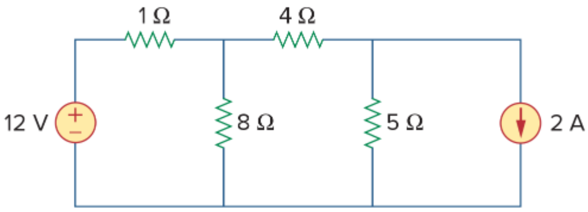

Figure 2.71
For **Prob. 2.7**.